

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) |  |  | double bond splits / opens / breaks (1)<br><br>the (vinyl chloride) molecules join / link together / form chains (1)<br><br>accept 'they join together'<br><br>$\left[ \begin{array}{cc} \text{H} & \text{Cl} \\   &   \\ -\text{C} & -\text{C}- \\   &   \\ \text{H} & \text{H} \end{array} \right] \quad (1)$<br><br>neutral answer – reference to heat / catalyst         | 3               |     |     | 3     |       |      |
|          | (c) |  |  | award (1) for any <b>two</b> of following<br><br><ul style="list-style-type: none"><li>• does not rust / corrode</li><li>• flexible / easy to shape</li><li>• lightweight / low density</li></ul><br>neutral answers<br>weather resistant<br>doesn't need to be painted<br>reference to cost<br><br>do not accept<br>low melting point<br>doesn't conduct electricity / heat | 1               |     |     | 1     |       |      |
|          |     |  |  | Question 3 total                                                                                                                                                                                                                                                                                                                                                             | 4               | 3   | 3   | 10    | 0     | 2    |